

EXERCISE NO.2

- Q1 List all distinct departments in the student table
SELECT department
FROM students_table.

Departmentlist
IT
HR
FINANCE

- Q2. Get the average age of students per department.
SELECT studentid, age, AVG(
FROM students_table
GROUP BY department

AVERAGE AGE OF STUDENTS PER DEPT		
HA	DEPARTMENT	AGE
	IT	31.5
	HR	33
	FINANCE	23.

- Q3 Show departments with more than 1 student.
SELECT department, SUM(C) > 1 - students
FROM students_table.
GROUP BY students

Number of departments with more than 1 student.	
department	student-count
IT	2
HR	2

- Q4 Get all students whose age is between 21-23
SELECT student_id, name (21-23)
FROM student_table
GROUP BY age AS ASC.

Students between the ages 21-23.			
student_id	name	age	department
3	Charlie	22	IT
2	Bob	22	HR
5	EVE	22	FINANCE ^{HR}
4	Diana	23	FINANCE

- Q5 List all students in the IT or HR department older than 21
SELECT * student_name, id
FROM student_table
WHERE IT = department; HR = department; > 21 ASC

List of students older than 21 in IT or HR dept			
ID	name	age	department
2	Bob	22	HR
5	EVE	22	HR
4	Diana	23	FINANCE

Q6. Show total credits per department, only for department with more than 5 total credits.

SELECT credits, department
FROM student table.

GROUP BY department; total credits.

TOTAL CREDITS PER DEPT	
DEPARTMENT	TOTAL CREDIT
IT	11
FINANCE	2
HR	3

Q7. List all courses that do not have 4 credits.

SELECT courses, department
FROM courses_table

WHERE courses NOT = 4 credits.

COURSES THAT DO NOT HAVE 4 CREDITS			
COURSE ID	COURSE NAME	DEPARTMENT	CREDITS
101	PYTHON SQL	IT	3
102	EXCEL DATA SCIENCE	FINANCE	2
104	STATISTICS	HR	3

Q8. Show the top 3 courses by credits in descending order.

SELECT courses, credits
FROM courses_table

WHERE top 3 credits

GROUP BY DESC

TOP 3 COURSES BY CREDITS			
course id	NAME COURSE	DEPT	credits
103	Data Science	IT	4
102	Python STATISTICS	IT	4
105	HR	HR	3

- Q9 Maximum, minimum & average grade across all enrollments.
SELECT enrollments, grade
FROM enrollments table
Group By maximum(), min(), average(), grade

MAX, MIN, AVERAGE		
MAX GRADE	MIN GRADE	AVERAGE GRADE

Q10.

MAX, MIN, AVERAGE ACROSS ALL ENROLLMENTS.		
MAX	MIN	AVERAGE
	CANCELLED	

Q10.

SELECT sum()
FROM enrollment
table.

group By number
of
sum

NO. OF ENROLLMENTS	
course id	enrollment count.
101	1
102	1
103	1
104	1
105	1

- Q11 Find total salary and total bonus per department.
SELECT SUM(total salary), SUM(total bonus)
FROM salary table
GROUP BY department.

total salary and bonus per dept		
department	total salary	total bonus
IT	12200	10500
HR	109000	7500
FINANCE	70000	6000

- Q12 Show departments where average salary is above 55000
SELECT department, salary (avg)
FROM salaries table
WHERE salary > 55000
ORDER BY > 55000

DEPARTMENTS WITH AVERAGE SALARY > 55000	
Department	AVERAGE SALARY
HR	89000
IT	76000
FINANCE	70000

- Q13 List of employees whose salary plus bonus is above 55000
 SELECT count(salary), sum(bonus)
 FROM salaries_table
 GROUP BY salary, bonus, > 55000

EMPLOYEES WITH SALARY & BONUS > 55000				
employee id	name	Salary	Bonus	TOTAL COMPENSATION
1	TOM	60000	5000	65000
2	JERRY	55000	4000	59000
3	SAKE	62000	5500	67500
5	TYKE	54000	3500	57500
4	SPIKE	70000	6000	76000

- Q14 Show total and average budget per department. Only include departments with average budget above 70000
~~set~~

TOTAL AND AVERAGE BUDGET PER DEPT > 70000		
Department	total budget	avrg budget
IT	270000	195000
FINANCE	80000	80000

SELECT department, total_budget,
 FROM projects_table
 GROUP BY (AVG) > 70000

Q15 List all projects with budgets between 50000 and 120000
excluding the Marketing department

SELECT projectid, projectname, department, budget
FROM projects table

GROUP BY ^{department} between 50000 and 120000; exclude marketing department

Projects with Budgets B/w 50000 and 120000 excl. marketing department			
project id	project name	department	budget
1			120 000
2	ATM Roll System	FINANCE	80 000
3			150 000
5	HR Portal	HR	50 000

END OF ASSIGNMENT.